

PAYSON, AZ, USA

WMO: 723745

Lat: 34.257N

Lon: 111.339W

Elev: 1572

StdP: 83.81

Time Zone: -7.00 (W07)

Period: 96-19

WBAN: 00374

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.3	-3.7	-22.3	0.6	7.0	-19.8	0.8	8.7	10.0	3.6	8.3	2.3	2.2	30	0.431

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	34.8	15.6	33.0	15.5	32.3	15.4	19.3	26.0	18.7	25.8	18.2	25.8	4.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	15.5	20.2	17.2	14.9	20.4	16.1	13.9	20.8	62.3	26.2	60.1	25.4	58.1	25.4	22.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.9	6.8	5.7	-8.9	37.1	2.9	1.2	-10.9	38.0	-12.6	38.8	-14.2	39.5	-16.3	40.4		
			-10.1	20.4	2.6	0.7	-11.9	20.9	-13.5	21.3	-14.9	21.7	-16.8	22.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.0	5.9	6.7	10.4	13.6	16.7	23.8	25.2	24.2	21.5	16.0	10.1	5.9
	DBStd	7.82	3.88	4.58	3.57	3.49	3.81	2.86	2.00	2.00	2.65	3.44	3.98	4.02
	HDD10.0	477	133	108	38	9	3	0	0	0	4	47	135	
	HDD18.3	1900	387	325	246	144	76	1	0	6	84	246	386	
	CDD10.0	2313	4	17	51	118	209	413	470	344	190	51	7	
	CDD18.3	696	0	0	0	3	24	164	212	181	100	12	0	0
	CDH23.3	7668	0	0	6	75	411	2244	2116	1746	922	147	2	0
	CDH26.7	3220	0	0	0	6	101	1086	977	736	296	19	0	0
Wind	WSAvg	2.1	2.0	2.0	2.2	2.6	2.7	2.5	2.1	1.9	1.9	2.0	1.9	1.7
Precipitation	PrecAvg	548	53	48	59	27	14	9	60	78	53	45	47	55
	PrecMax	830	177	230	190	114	48	73	109	185	171	262	135	239
	PrecMin	378	0	0	0	0	0	0	19	18	0	0	0	0
	PrecStd	128	44	49	47	29	15	15	25	49	44	49	35	59
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.8	21.2	24.0	27.4	31.3	37.4	36.0	35.0	32.5	28.9	23.0	18.0
		MCWB	6.5	8.2	9.7	10.3	13.0	15.4	16.9	16.0	15.8	12.7	9.0	6.4
	2%	DB	16.2	19.1	22.2	25.8	29.1	35.1	34.0	32.9	31.3	26.4	21.2	16.4
		MCWB	5.5	7.6	9.0	9.7	12.1	15.0	16.3	16.5	15.2	11.5	8.6	5.9
	5%	DB	14.2	17.3	20.0	23.7	27.5	33.7	32.8	32.3	30.0	25.0	19.1	14.2
		MCWB	4.8	6.6	7.8	9.0	10.9	14.2	16.4	16.5	14.5	11.1	8.0	5.6
	10%	DB	12.6	14.8	17.9	22.2	26.0	32.4	32.1	31.0	28.0	22.9	17.4	12.4
		MCWB	4.2	5.6	6.7	8.4	10.2	13.8	16.3	16.5	14.7	10.6	7.4	4.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.5	10.2	10.6	11.7	14.4	18.1	20.2	19.7	19.3	16.0	11.0	9.9
		MCDB	15.0	18.0	21.6	23.2	29.5	30.1	25.5	26.6	25.2	18.2	18.1	13.3
	2%	WB	7.0	8.4	9.5	10.5	13.0	16.8	19.6	19.0	18.0	14.5	10.0	8.1
		MCDB	12.1	16.5	20.3	23.4	27.1	30.3	25.9	26.3	24.7	20.0	17.6	12.7
	5%	WB	6.0	7.3	8.5	9.6	11.7	16.0	19.0	18.5	17.3	13.4	9.0	6.9
		MCDB	11.7	15.4	17.7	21.8	25.4	29.7	25.8	26.1	24.6	20.7	16.9	12.0
	10%	WB	5.1	6.3	7.5	8.8	10.7	15.1	18.5	18.0	16.6	12.3	8.2	6.0
		MCDB	11.5	12.7	16.6	20.5	24.4	29.1	25.6	26.1	24.3	20.2	15.8	10.9
Mean Daily Temperature Range	5% DB	MDBR	10.8	10.6	12.6	13.3	14.2	15.4	12.7	12.6	12.4	12.1	11.1	9.4
		MCDBR	13.1	13.9	14.7	15.6	17.0	16.8	14.2	14.6	14.5	14.1	12.9	11.8
		MCWBR	6.8	7.0	6.7	6.9	6.7	6.1	4.6	4.4	5.0	6.1	6.3	6.5
	5% WB	MCDBR	10.5	11.9	13.2	14.4	14.8	14.0	11.2	11.8	11.2	10.7	11.2	10.1
		MCWBR	6.2	6.5	6.4	6.5	6.3	5.4	3.7	4.1	4.6	5.3	5.9	5.9
Clear-Sky Solar Irradiance	taub	0.246	0.251	0.273	0.286	0.297	0.302	0.354	0.350	0.304	0.283	0.253	0.246	
	taud	2.680	2.683	2.594	2.545	2.556	2.538	2.494	2.484	2.610	2.619	2.671	2.681	
	Ebn at Noon	995	1019	1010	1002	986	975	923	924	961	964	976	976	
	Edh at Noon	69	77	92	101	102	103	108	107	90	81	69	65	
All-Sky Solar Radiation	RadAvg	3.25	4.18	5.69	7.11	8.00	8.42	6.89	6.41	5.93	4.80	3.67	2.92	
	RadStd	0.30	0.38	0.37	0.30	0.45	0.38	0.42	0.31	0.24	0.36	0.20	0.19	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	+0.35	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+81	+33

Nomenclature: See separate page